Rocket Equation

title: “Session 4: Reproducible Documents” author: Kira Gruhlke format: html: code-fold: true pdf: echo: false pdf-engine: pdflatex toc: true number-depth: 2 number-sections: true papersize: a4 documentclass: article jupyter: python3

```
# the first cell provides metadata (raw cell)
```

At time t_0 to rocket starts to expel gas at a *constant mass flow rate* R meassured in kg/s and *exhaust velocity relative to the rocket* v_e in m/s.

$$\frac{dv}{dt} = -\frac{F}{m(t)} = -\frac{Rv_e}{m(t)} \tag{1}$$

Integrating both sides of Equation 1 from 0 to T we get

```
# in between the $$ in Markdown, there is a maths-term
# backslash frac gives a fraction
```

Reading a Data file and plotting

Copy from session 3 to use for session 4

Import needed files

```
import numpy # import needed files first
from matplotlib import pyplot #import from matplotlib the function for this code
```

Open the files and read it, so it is usable

```
#create lists
time= []
absorption=[]
```

```
dataFile=open("data/pulse_data.csv")
```

```
!pwd #where am I
```

```
/home/kigr11/Spacemed-2026/kigr11
```

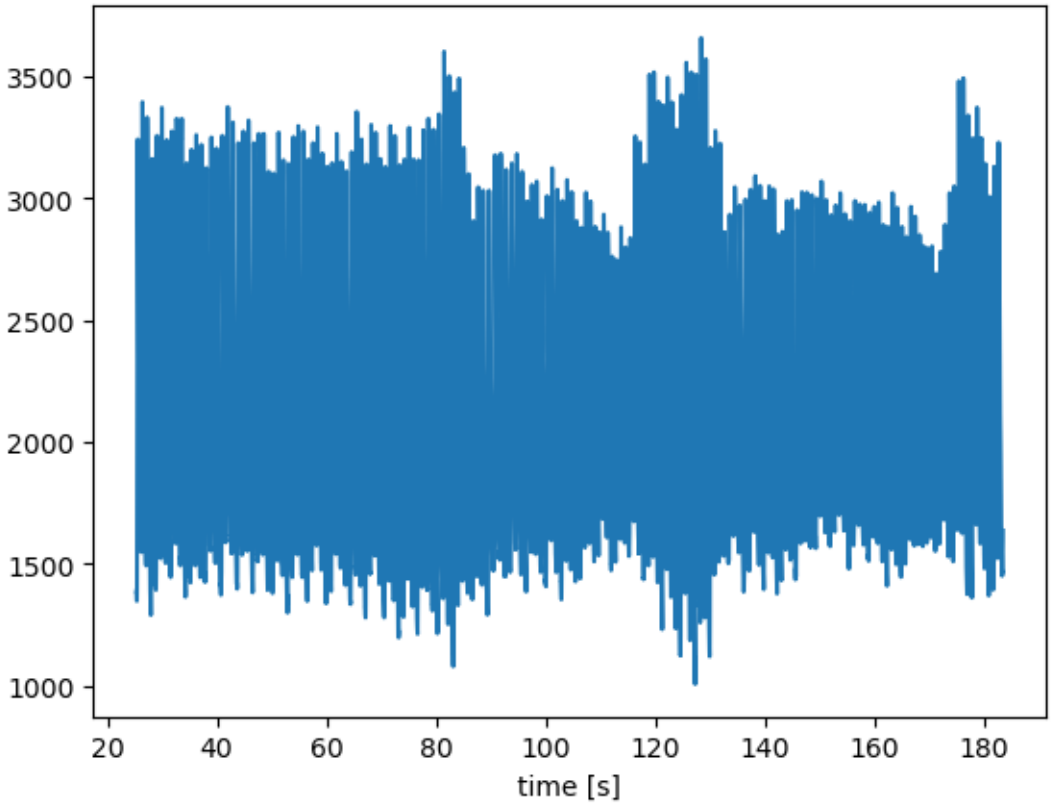
```
dataFile.readline() # if we run this, we read a line and then discard it, so jump to the nex
for line in dataFile.readlines() : # now the for loop starts with the second line
    line=line.split(",") #split at the comma
    time.append(float(line[0])) # 0 = before the comma; float because it would have been a s
    absorption.append(float(line[1])) # 1 = after the comma
# if I reexecude the loop later, I might already be in the wrong line of the file
```

Basic Plot

```
pyplot.plot(time, absorption)
pyplot.xlabel("time [s]")
```

```
Text(0.5, 0, 'time [s]')
```

(a) Measurements from an oxymetry sensor as a function of time for 3 cycles



(b)

Figure 1